Design Patterns

Advanced Programming

Brent van Bladel

Design Principles

- Liskov substitution principle
- Type Conformity
- Principle of Closed Behavior

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- Open–closed principle
- Low coupling / high cohesion

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- Type Conformity
- Principle of Closed Behavior
- Single-responsibility principle
- Open–closed principle
- Low coupling / high cohesion

Object-oriented designs following these principles are more understandable, flexible, and maintainable.

Single-responsibility principle

“Every class must have only one responsibility.”

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“Every class must have only one responsibility.”

```
1.  class FileManager {
2.   public:
3.     void write(string filename, string text);
4.     string read(string filename);
5.  };
```

Violates SRP

Single-responsibility principle

"Every class must have only one responsibility."

```
1.  class FileManager {
2.   public:
3.     void write(string filename, string text);
4.     string read(string filename);
5.  };
```

Violates SRP

```
1.  class FileWriter {
2.    public:
3.      void write(string filename, string text);
4.  };
5.
6.  class FileReader {
7.    public:
8.      string read(string filename);
9.  };
10.
11. class FileManager
12.     : public FileWriter,  public FileReader {};
```

Follows SRP

Open–closed principle

“Software entities should be open for extension, but closed for modification”

Open–closed principle

"Software entities should be open for extension, but closed for modification"

```
1.  class FileReader {
2.   XMLReader _xmlReader;
3.   JSONReader _jsonReader;
4.  public:
5.   string read(string filename) {
6.     if (endsIn(filename, "XML")
7.       return _xmlReader.readXML(filename);
8.     else if (endsIn(filename, "JSON")
9.       return _jsonReader.readJSON(filename);
10.   };
11. };
```

Violates OCP

Open–closed principle

"Software entities should be open for extension, but closed for modification"

```
1.  class FileReader {
2.   XMLReader _xmlReader;
3.   JSONReader _jsonReader;
4.  public:
5.   string read(string filename) {
6.     if (endsIn(filename, "XML")
7.       return _xmlReader.readXML(filename);
8.     else if (endsIn(filename, "JSON")
9.       return _jsonReader.readJSON(filename);
10.  };
11. };
```

Violates OCP

```
1.  class FileReader {
2.  public:
3.    virtual string read(string filename);
4.  };
5.  class XMLReader : public FileReader {
6.  public:
7.    string read(string filename) override;
8.  };
9.  class JSONReader : public FileReader {
10. public:
11.   string read(string filename) override;
12. };
```

Follows OCP

Low coupling / high cohesion

Coupling is the degree to which the different classes depend on each other.

→ Should be as low as possible.

Cohesion is the degree to which the elements of a class belong together.

→ Should be as high as possible.

Design Pattern

A **design pattern** provides a typical solution to a common problem in software design.

- Tried and tested solutions to common problems.
- Follow design principles.
- Define a common language that developers can use to communicate more efficiently.

Design Pattern

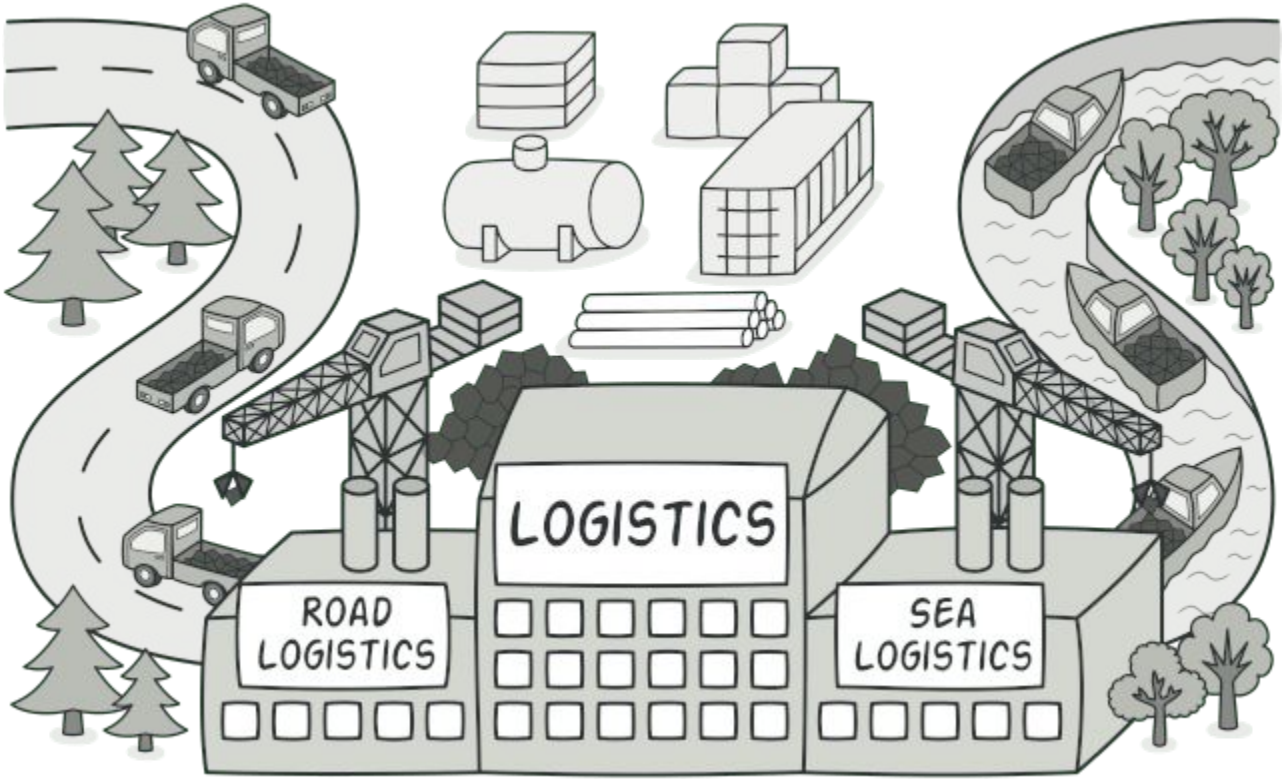
For each pattern, we will examine:

- the **problem** the pattern tries to solve.
- the **solution** that solves the problem.
- the **structure** which shows the relation between the different parts of the pattern.
- the **benefits** and the **drawbacks** of the pattern.

Creational Patterns

These patterns provide various object creation mechanisms, which increase flexibility and reuse of existing code.

Factory Method



Factory Method: Problem

```
1.  if (request == "Circle") {
2.      Circle c();
3.      cout << c.area();
4.  } else if (request == "Rectangle") {
5.      Rectangle r();
6.      cout << r.area();
7.  }
```

- Given some variable, we want to construct a certain object.
  → We end up with many conditionals in our code that switch behavior depending on the class.
- What if we need to add a new class?

Factory Method: Solution

Factory Method

```
1.  Shape* shapeFactory(string request) {
2.   if (request == "Circle")
3.     return new Circle();
4.   if (request == "Rectangle")
5.     return new Rectangle();
6.   return nullptr;
7.  }
```

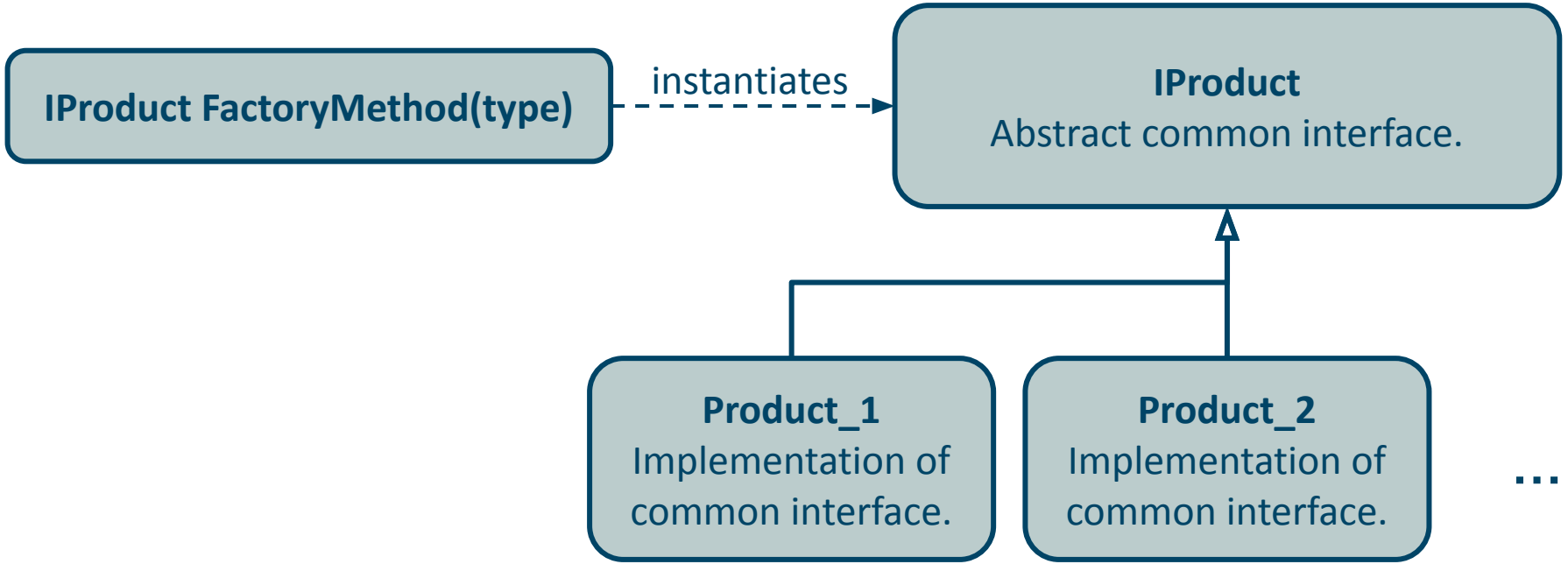
```
1.  ...
2.  Shape* s = shapeFactory(request);
3.  cout << s->area();
4.  ...
```

Common interface

```
1.  class Shape{
2.  public:
3.    virtual float area() const = 0;
4.  };
5.  class Circle: public Shape{
6.  public:
7.    float area() const override;
8.  };
9.  class Rectangle: public Shape{
10. public:
11.   float area() const override;
12. };
```

Replace direct object construction calls with calls to a special factory method.

Factory Method: Structure



Factory Method: Benefits & Drawbacks

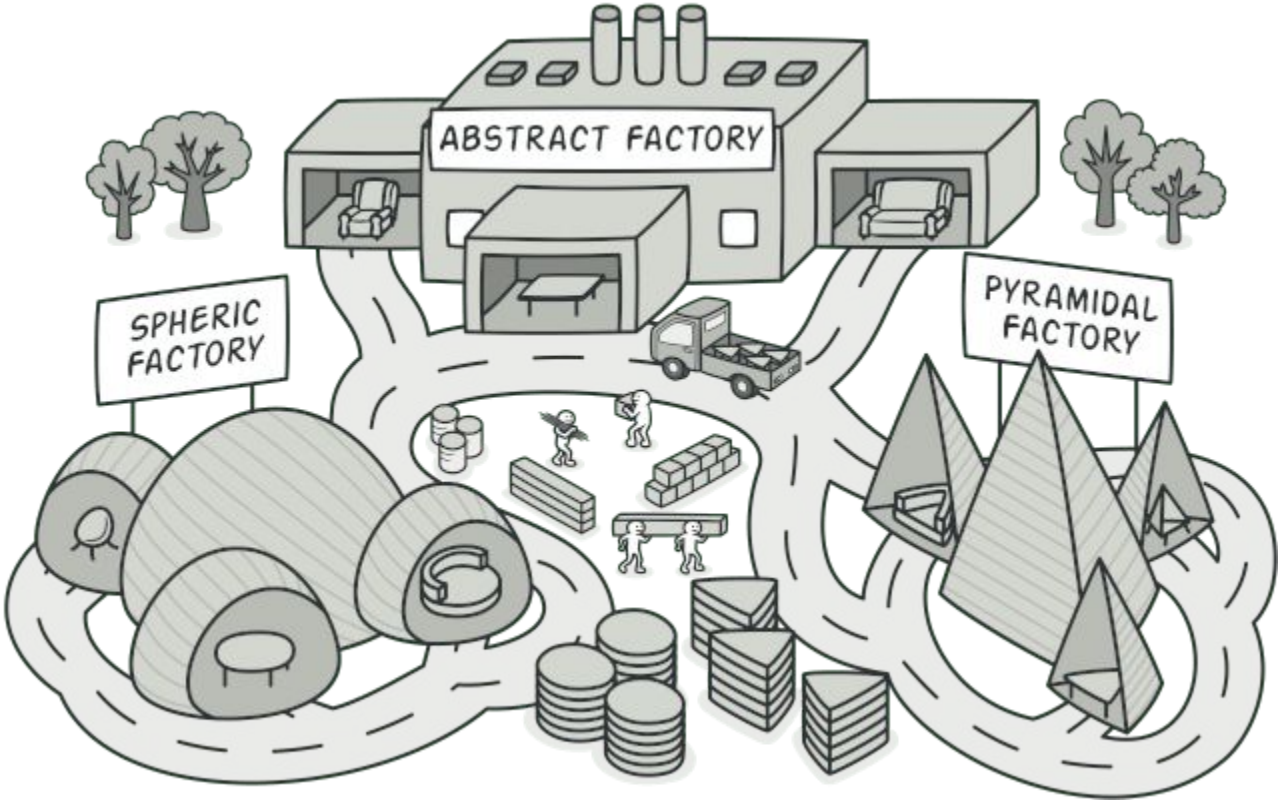
Benefits

- Avoid close coupling between the creator and the concrete products.
- **Single Responsibility Principle**. You can move the product creation code into one place in the program, making the code easier to support.
- **Open/Closed Principle**. You can introduce new types of products into the program without breaking existing client code.

Drawbacks

- The code may become more complicated since you need to introduce a lot of new subclasses.

Abstract Factory



Abstract Factory: Problem

```
1.  Shape* shapeFactory(string shape, string type) {
2.   if (request == "Rectangle") {
3.     if (type == "size")
4.       return SizedRectangle();
5.     if (type == "coordinates")
6.       return CoordinatedRectangle();
7.   } else if (...) {
8.     ...
```

```
1.  class SizedRectangle: public Rectangle {
2.      float _width, _length;
3.    public:
4.      float area() const override;
5.  };
6.  class CoordinatedRectangle:
7.                                  public Rectangle {
8.      vector<Coordinate> _corners;
9.    public:
10.     float area() const override;
11. };
```

- We want to construct certain objects via a factory method, but each object has multiple implementations.
- How can we guarantee that the constructed objects work together?

Abstract Factory: Solution

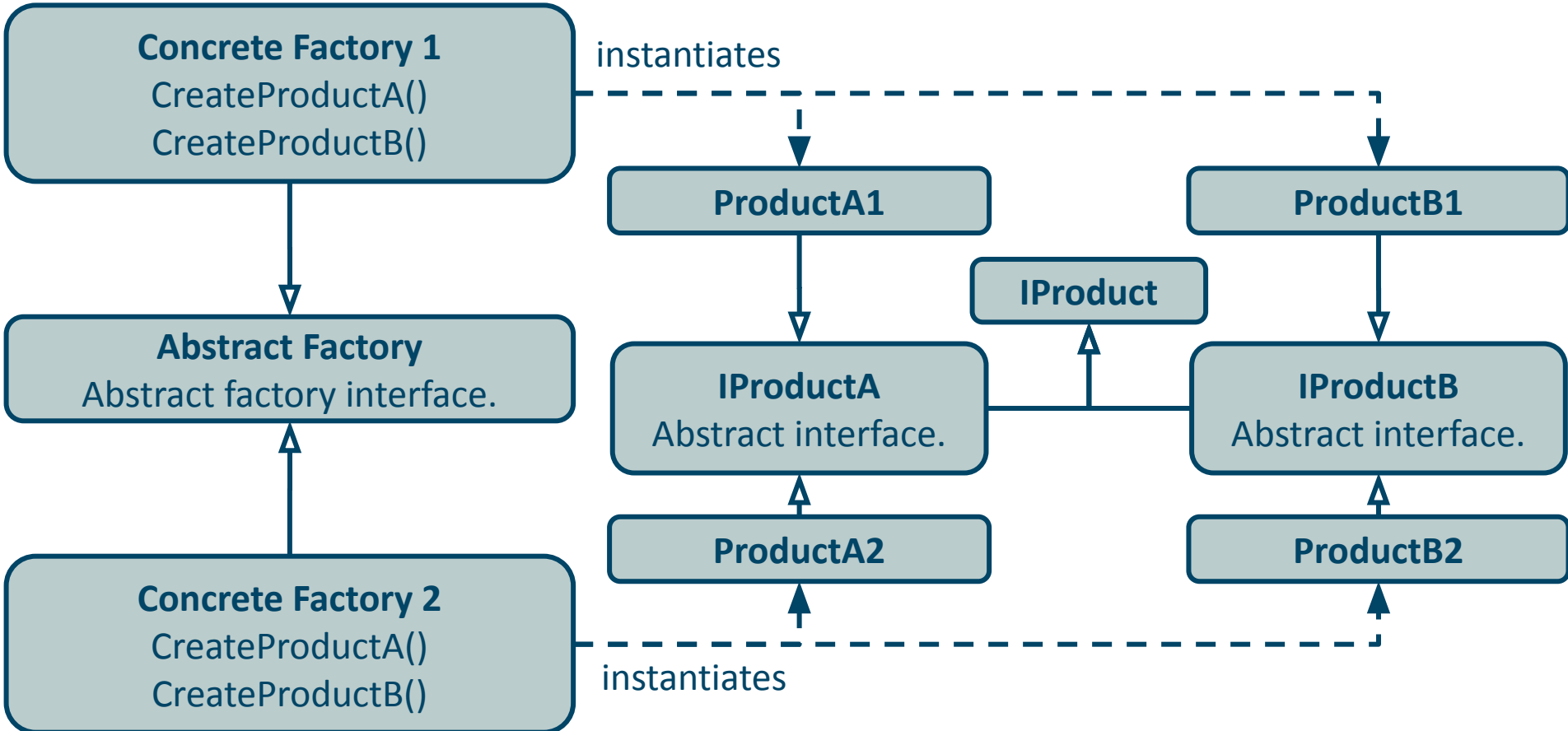
```
1.  ShapeFactory* f = SizedShapeFactory();
2.
3.  Circle* c = f->createCircle();
4.  Rectangle* r = f->createRectangle();
```

Place the factory methods in a class that implement an abstract factory interface.

Abstract Factory

```
1.  class ShapeFactory {
2.   public:
3.     virtual Shape* createRectangle() = 0;
4.     virtual Shape* createCircle() = 0;
5.  };
6.  class SizedShapeFactory: public ShapeFactory {
7.  public:
8.     Shape* createRectangle() override;
9.     Shape* createCircle() override;
10. };
11. class CoordinatedShapeFactory:
12.                                  public ShapeFactory {
13. public:
14.    Shape* createRectangle() override;
15.    Shape* createCircle() override;
16. };
```

Abstract Factory: Structure



Abstract Factory: Benefits & Drawbacks

Benefits

- Guarantee that the created products from a factory are compatible.
- Avoid close coupling between the creator and the concrete products.
- **Single Responsibility Principle**. You can move the product creation code into one place in the program, making the code easier to support.
- **Open/Closed Principle**. You can introduce new types of products into the program without breaking existing client code.

Drawbacks

- The code may become more complicated since a lot of new interfaces and classes are introduced.

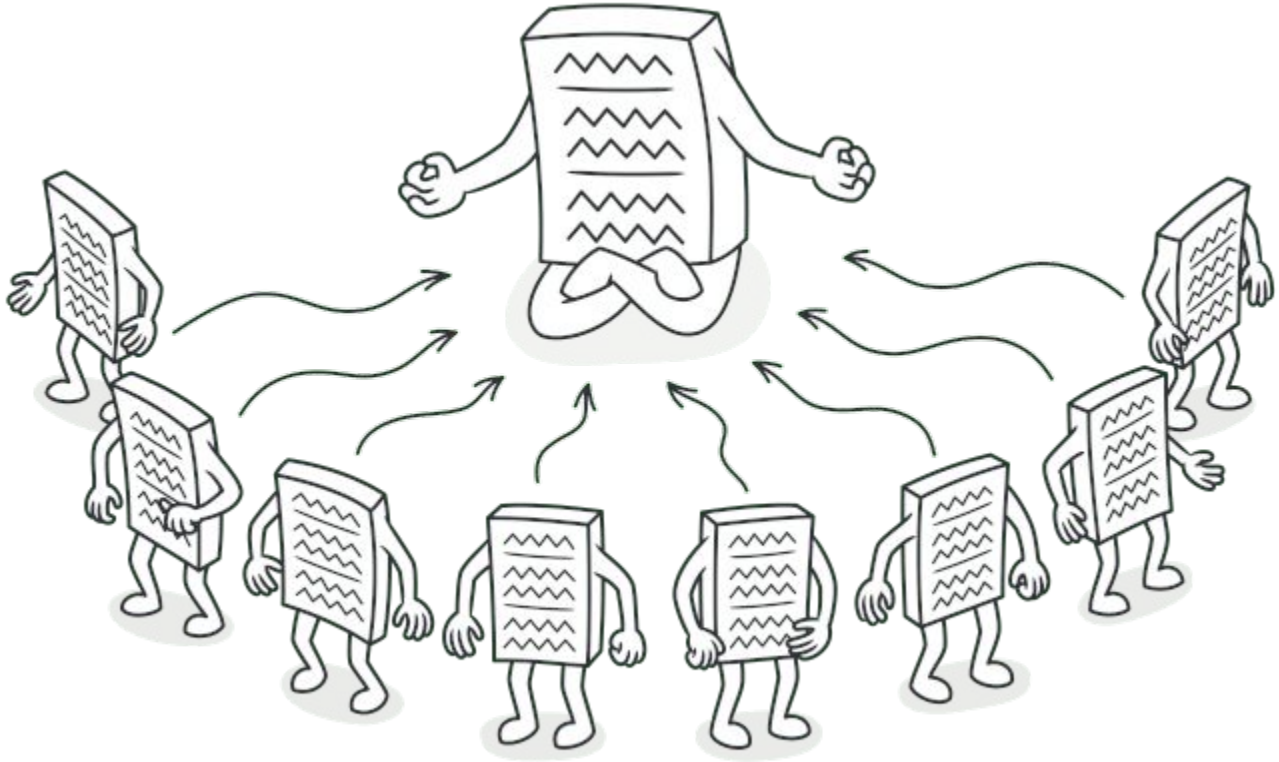
Combining Design Patterns

```
1.  class ShapeFactory {
2.   public:
3.    virtual Shape* createShape(string request) = 0;
4.  };
5.
6.  class SizedShapeFactory: public ShapeFactory {
7.   public:
8.     Shape* createShape(string request) override;
9.  };
10.
11. class CoordinatedShapeFactory: public ShapeFactory {
12.  public:
13.    Shape* createShape(string request) override;
14. };
```

Abstract Factory

Factory Method

Singleton



Singleton: Problem

```
1.  ...
2.  ShapeFactory* f1 = SizedShapeFactory();
3.  Shape* s1 = shapeFactory->createRectangle();
4.  ...
5.  ShapeFactory* f2 = CoordinatedShapeFactory();
6.  Shape* s2 = shapeFactory->createCircle();
7.  ...
```

- We want to have exactly one object of a certain class in our program.
- How can we guarantee that only one object is created?

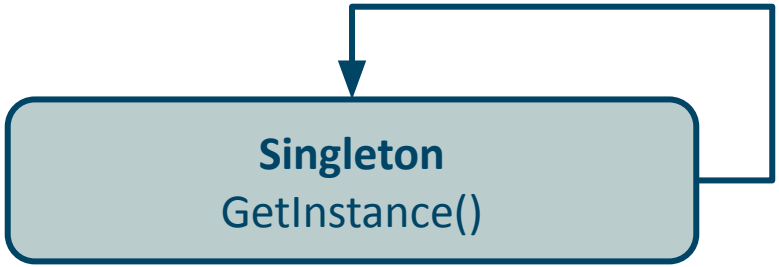
Singleton: Solution

```
1.  class ShapeFactory {
2.    private:
3.      static ShapeFactory* _instance;
4.      ShapeFactory() = default;
5.    public:
6.      ~ShapeFactory() = default;
7.
8.      static ShapeFactory* getInstance() {
9.        if (!_instance)
10.         _instance = new ShapeFactory();
11.       return _instance;
12.     }
13.
14.     Shape* createShape(string request);
15. };
```

1) Make the constructor private. Have a private static member containing the singleton.

2) Create a static method that creates the singleton the first time it is called. All following calls to this method return the already created object.

Singleton: Structure



Singleton: Benefits & Drawbacks

Benefits

- Guarantee that a class has only one instance.
- Global access point to the instance.
- The singleton object is initialized only when it’s requested for the first time.

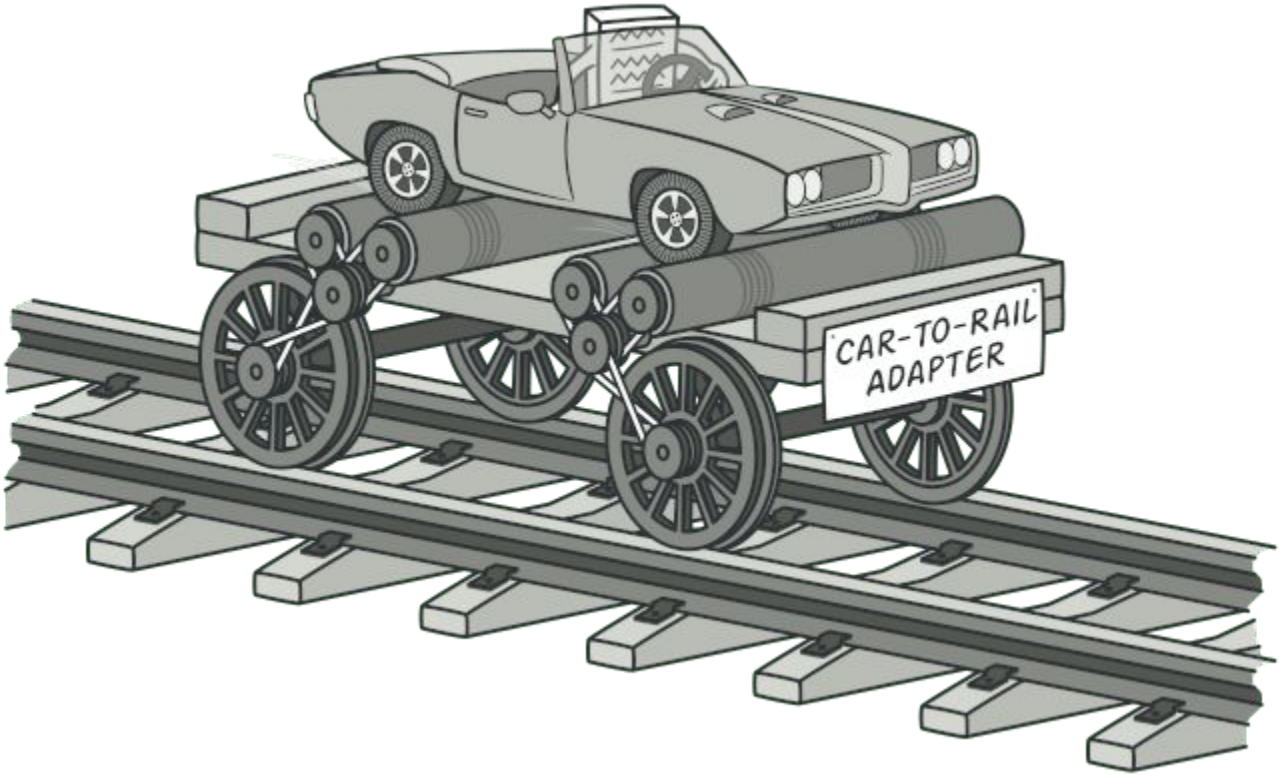
Drawbacks

- Violates the **Single Responsibility Principle**.
- Can cause a bad design. They hide the dependencies of your application, and they inherently cause code to be tightly coupled.
- Requires special treatment in a multithreaded environment.

Structural Patterns

These patterns explain how to assemble objects and classes into larger structures, while keeping these structures flexible and efficient.

Adapter



Adapter: Problem

```
1.  ...
2.  MyCircle c;
3.  LibSquare s;
4.
5.  cout << c.area() << endl;
6.  cout << s.calculateArea()
    << endl;
7.  ...
```

```
1.  class MyCircle : public MyShape {
2.    public:
3.      float area() override;
4.  };
5.
6.  class LibSquare : public LibShape{
7.    public:
8.      float calculateArea() override;
9.  };
```

- We want to use two similar classes with a different interface.
- How can we call these interchangeably?

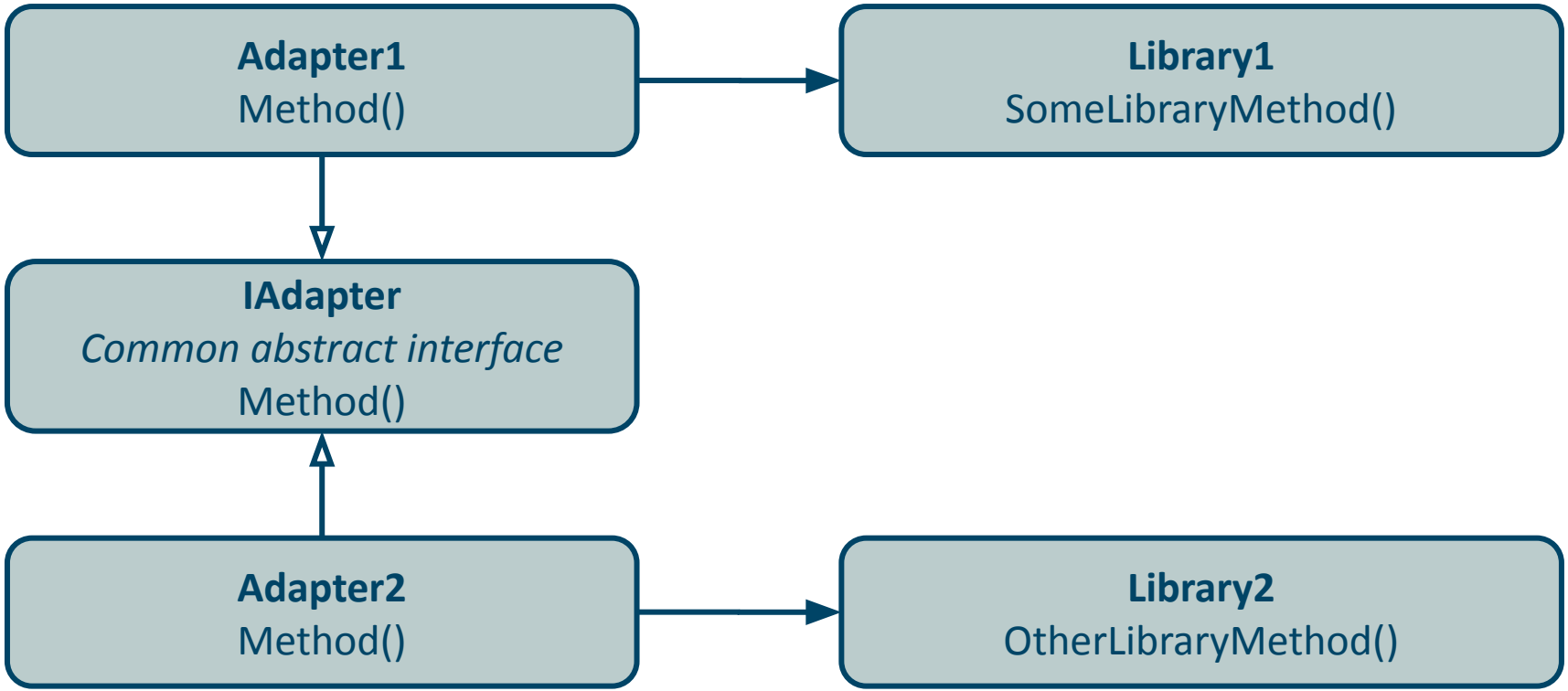
Adapter: Solution

```
1.  ...
2.  vector<MyShape *> shapes;
3.
4.  shapes.push_back(new MyCircle());
5.  shapes.push_back(new MyRectangleAdapter());
6.
7.  for (auto shape: shapes)
8.    cout << shape->area() << endl;
9.  ...
```

```
1.  class MySquareAdapter : public MyShape {
2.    private:
3.      LibSquare _square;
4.    public:
5.      float area() override {
6.          return _square.calculateArea();
7.      };
8.  };
```

Implement an adapter in one interface that wraps the other class

Adapter: Structure



Adapter: Benefits & Drawbacks

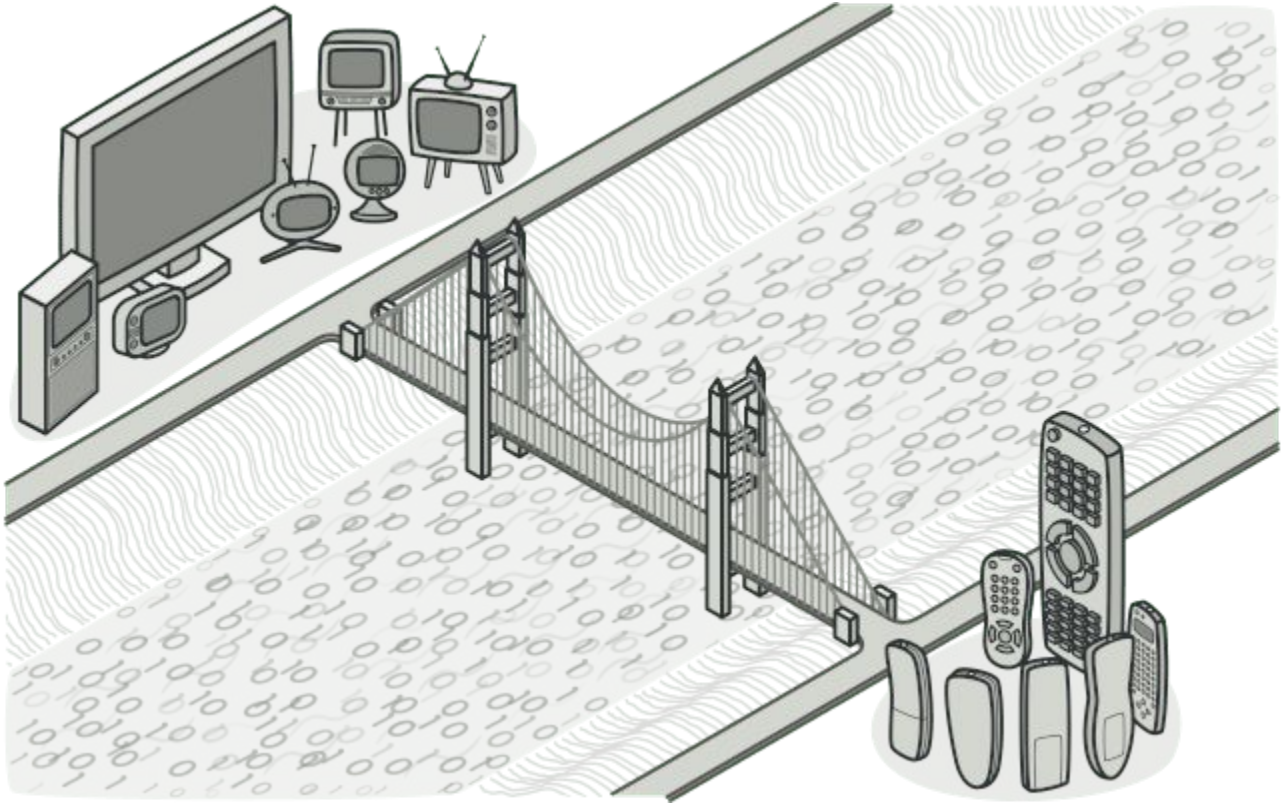
Benefits

- **Single Responsibility Principle**. Separate the interface or data conversion code from the primary business logic of the program.
- **Open/Closed Principle.** Introduces new types of adapters into the program without breaking the existing client code.

Drawbacks

- The overall complexity of the code increases because you need to introduce a set of new interfaces and classes.

Bridge



Bridge: Problem

```
1.  class Shape {
2.    public:
3.      virtual float area() = 0;
4.      virtual void printColor() = 0;
5.  };
6.
7.  class RedCircle : public Shape {
8.    public:
9.      float area() override;
10.     void printColor() override;
11. }
12. class BlueCircle : public Shape {
13.   public:
14.     float area() override;
15.     void printColor() override;
16. }
```

```
1.  class RedSquare : public Shape {
2.    public:
3.      float area() override;
4.      void printColor() override;
5.  }
6.
7.  class BlueSquare : public Shape {
8.    public:
9.      float area() override;
10.     void printColor() override;
11. }
```

- We want to specialize a class.
  Example: platform dependent classes.
- Adding a new class or specialization makes the hierarchy grow exponentially.

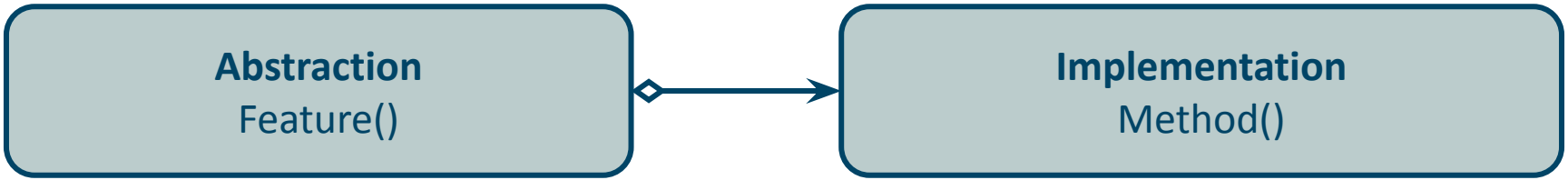
Bridge: Solution

```
1.  class Shape {
2.    private:
3.      Color _color;
4.    public:
5.      virtual float area() = 0;
6.      virtual void printColor() = 0;
7.  };
```

Switch from inheritance to object composition.

→ Extract one of the dimensions into a separate class hierarchy.

Bridge: Structure



Bridge: Benefits & Drawbacks

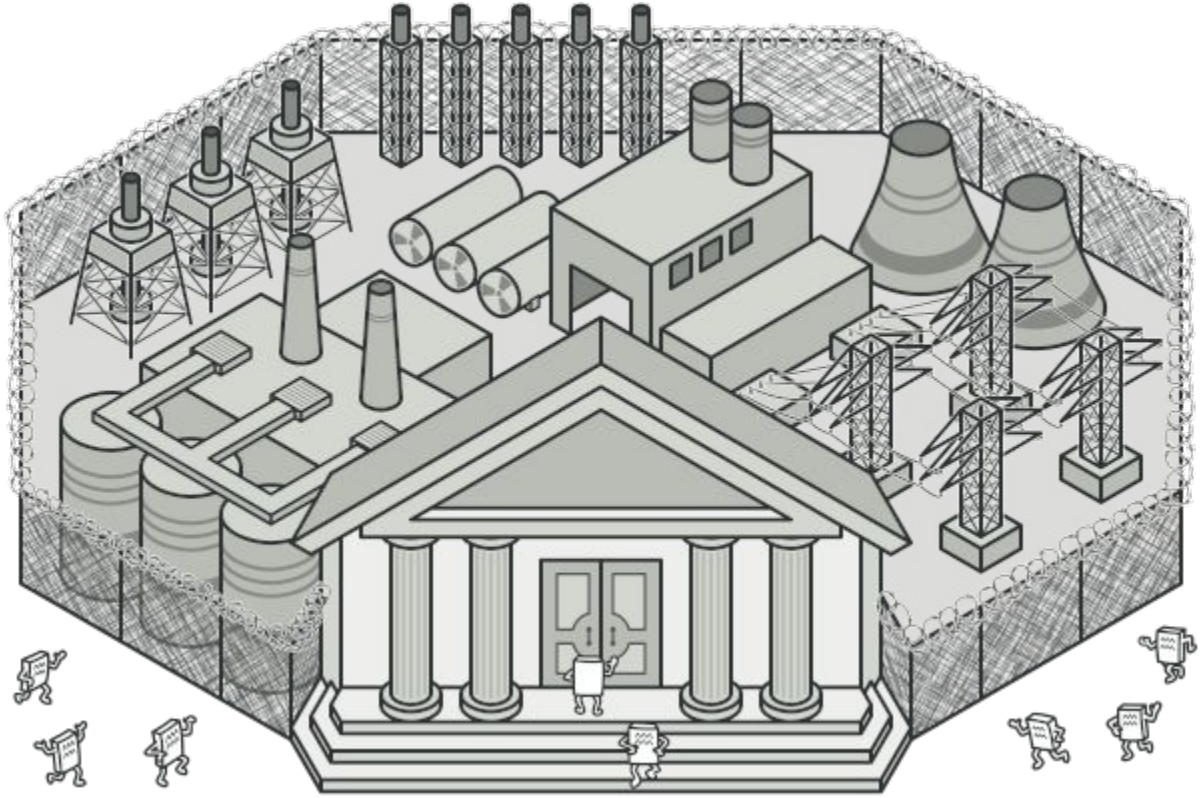
Benefits

- The caller works with high-level abstractions. It isn't exposed to the details.
- **Single Responsibility Principle**. Focus on high-level logic in the abstraction and on low-level details in the implementation.
- **Open/Closed Principle.** Introduce new abstractions and implementations independently from each other.

Drawbacks

- The overall complexity of the code can increase when applying the bridge pattern to a highly cohesive class.

Facade



Facade: Problem

```
1.  class FuelTank{
2.    public:
3.      Fuel* getFuel();
4.  };
5.  class Engine {
6.    public:
7.      void start();
8.      void consumeFuel();
9.      void chargeBattery();
10.     void drive();
11.     void stop();
12. };
13. class HeadLights {
14.   public:
15.     void consumeBattery();
16. };
```

- We want to use a bunch of classes that form a complex subsystem.
- Using this subsystem requires us to keep many objects and call specific functions in the correct order.

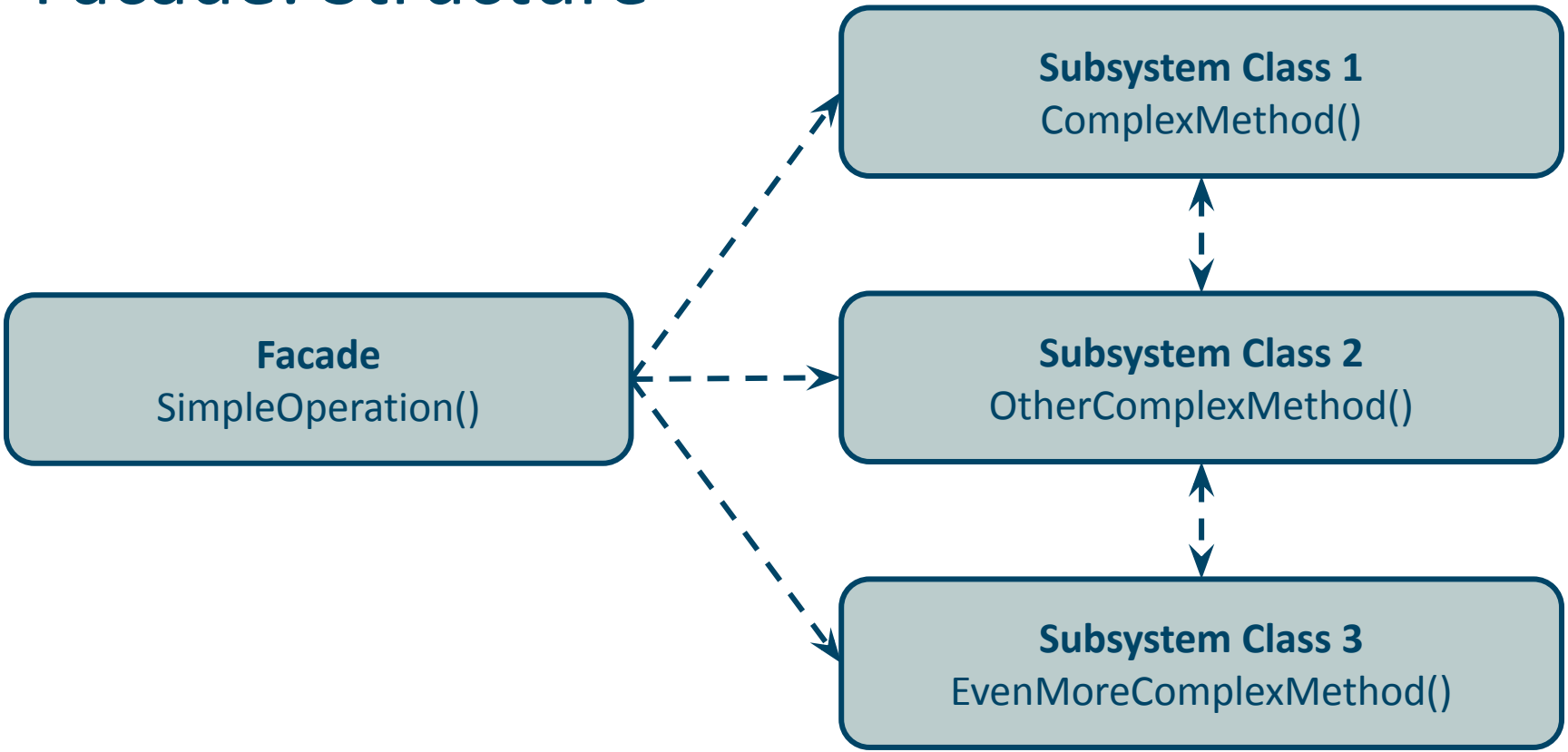
Facade: Solution

```
1.  class Car {
2.    private:
3.      FuelTank _tank;
4.      Engine _engine;
5.      HeadLights _lights;
6.    public:
7.      void drive();
8.  };
```

Implement a facade class that manages the objects and handles all the complex calls.

→ The user only needs to create the facade and call the simpler facade calls.

Facade: Structure



Facade: Benefits & Drawbacks

Benefits

- Isolate your code from the complexity of a subsystem.

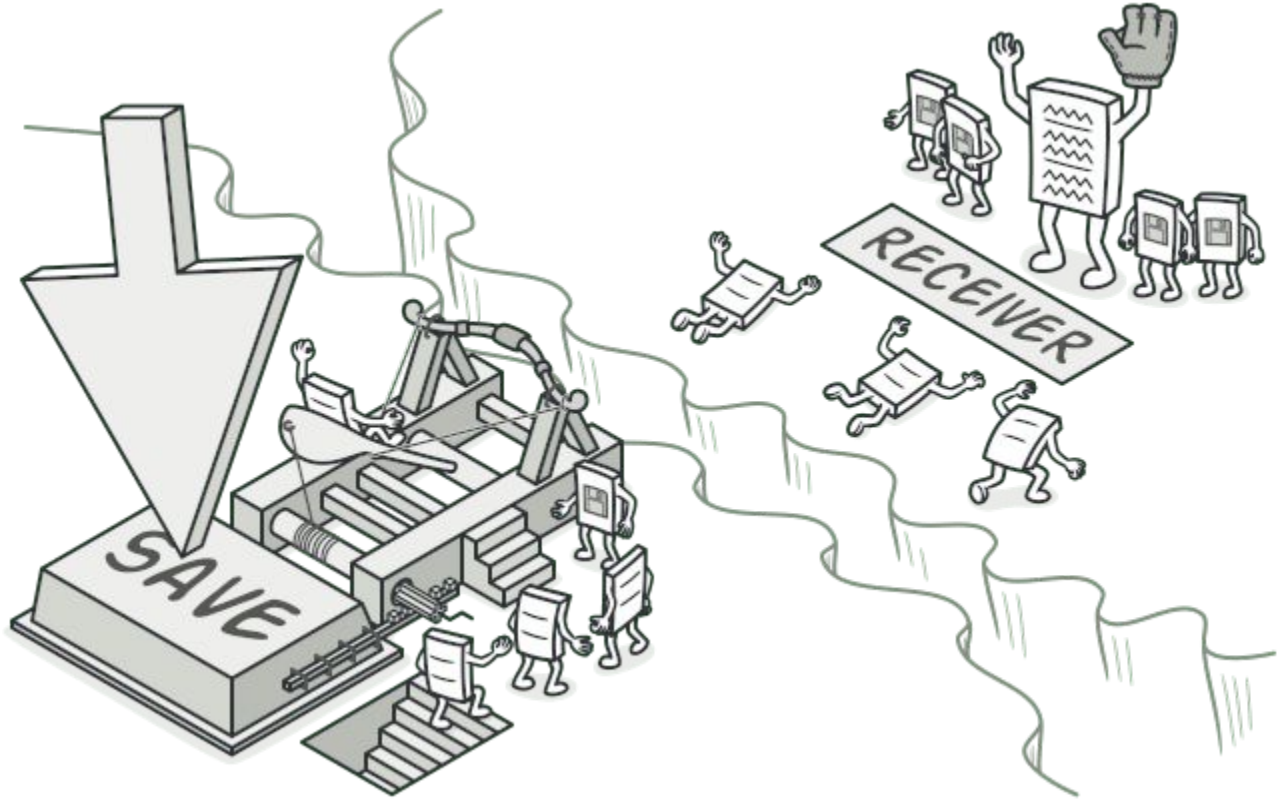
Drawbacks

- A facade can become a god object coupled to all classes of the subsystem.

Behavioral patterns

These patterns are concerned with algorithms and the assignment of responsibilities between objects.

Command



Command: Problem

```
1.  class LightOnButton : public Button {
2.    public:
3.      void press() override;
4.  };
5.  class LightOffButton : public Button {
6.    public:
7.      void press() override;
8.  };
9.  class SoundOnButton : public Button {
10.   public:
11.     void press() override;
12. };
13. class SoundOffButton : public Button {
14.   public:
15.     void press() override;
16. };
```

- We have a bunch of classes that each perform the same action on different targets.
- Adding a new target will require us to create many additional classes.

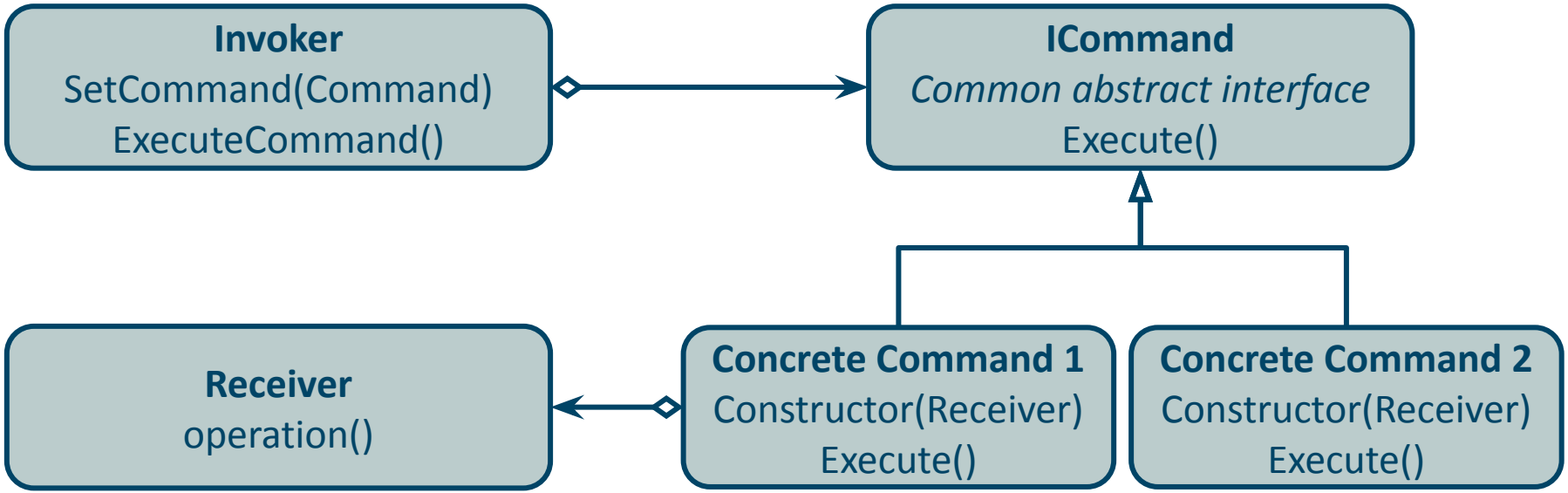
Command: Solution

Implement a class that represents the action.

→ Adding a new target does not increase the amount of classes.

```
1.  class Command {
2.    Device* _device;
3.  };
4.
5.  class TurnOnCommand: public Command {
6.    public:
7.      TurnOnCommand(Device* d) : _device{d} {  };
8.      void execute() override { _device->on(); };
9.  };
10. class TurnOffCommand: public Command {
11.   public:
12.     TurnOnCommand(Device* d) : _device{d} {   };
13.     void execute() override { _device->off(); };
14. };
15.
16. class Button {
17.   public:
18.     Command* press();
19. };
```

Command: Structure



Command: Benefits & Drawbacks

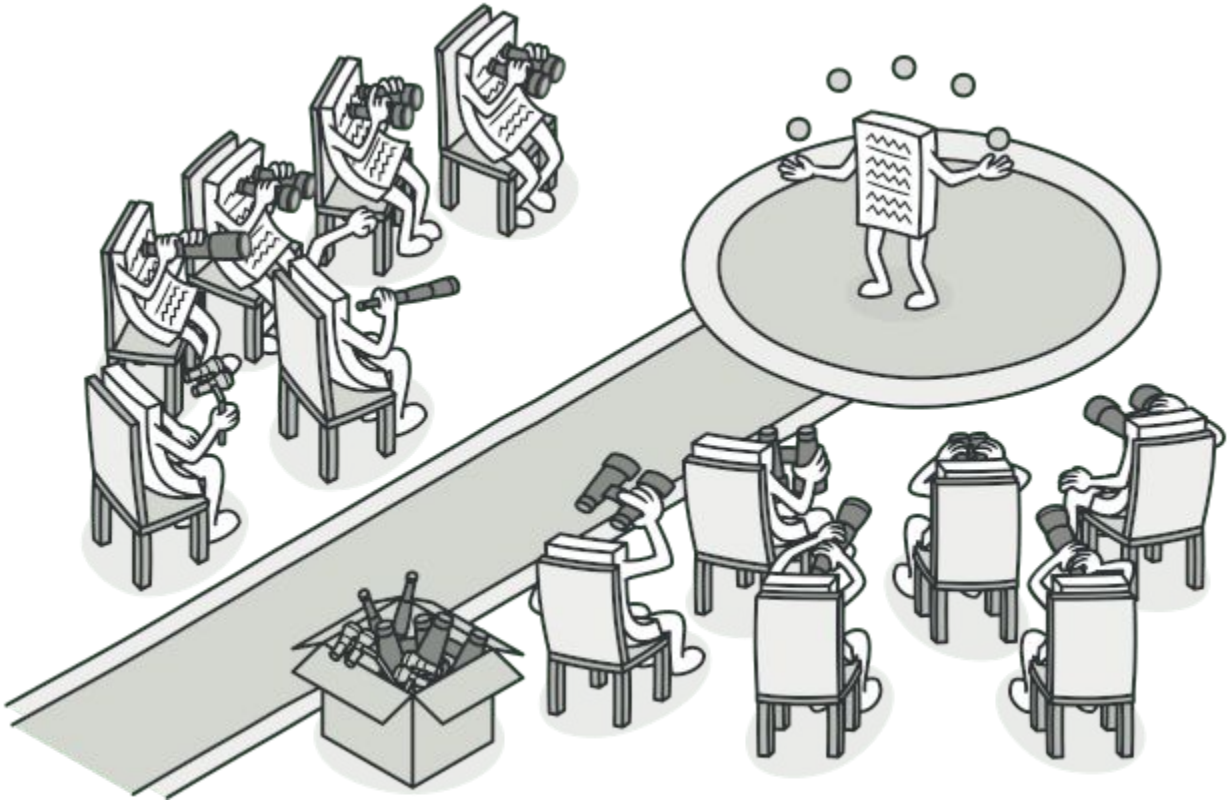
Benefits

- **Single Responsibility Principle**. Decouple the classes that invoke operations from the classes that perform these operations.
- **Open/Closed Principle.** Introduce new commands without breaking existing client code.
- Simple commands can be combined into a complex one.

Drawbacks

- The code may become more complicated since you're introducing a whole new layer between senders and receivers.

Observer



Observer: Problem

```
1.  class Weather {
2.      float _temperature;
3.      bool _isRaining;
4.    public:
5.      void setWeatherData(float t, bool r);
6.  };
7.
8.  class TemperatureDisplay {
9.    public:
10.     void display(float t);
11. };
12. class WaterCollector {
13.   public:
14.     void turnOn();
15.     void turnOff();
16. };
```

- We have classes that are interested in the state of an object and need to be updated when that state changes.
- What if we need to add a new class?

Observer: Solution

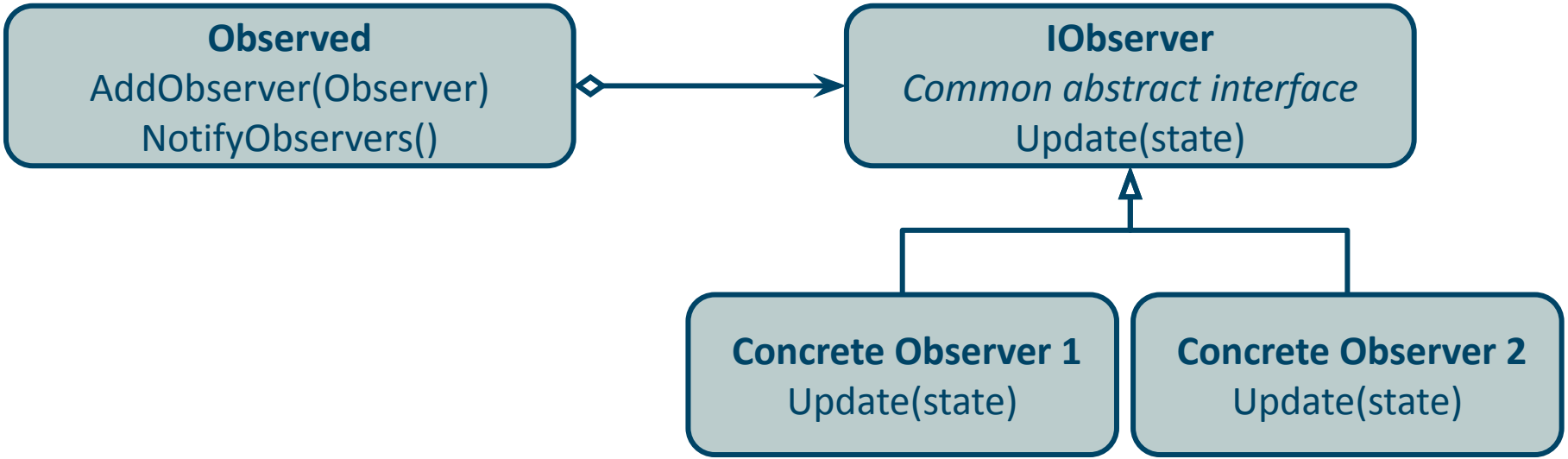
```
1.  class Weather {
2.      float _temperature;
3.      bool _isRaining;
4.      list<Observer*> _observers;
5.    public:
6.      void addObserver(Observer* o);
7.
8.      void setWeatherData(float t, bool r){
9.          _temperature = t;
10.         _isRaining = r;
11.         for (auto observer: _observers)
12.             observer.update(this);
13.     }
14. };
```

```
1.  class Observer {
2.    public:
3.      virtual void update(Weather* w) = 0;
4.  };
5.  class TemperatureDisplay : public Observer {
6.    public:
7.      void update(Weather* w) {
8.          display(w->getTemp(); };
9.      void display(float t);
10. };
```

Add a common interface to all interested classes, and store them in a list.

→ update all list items on state change

Observer: Structure



Observer: Benefits & Drawbacks

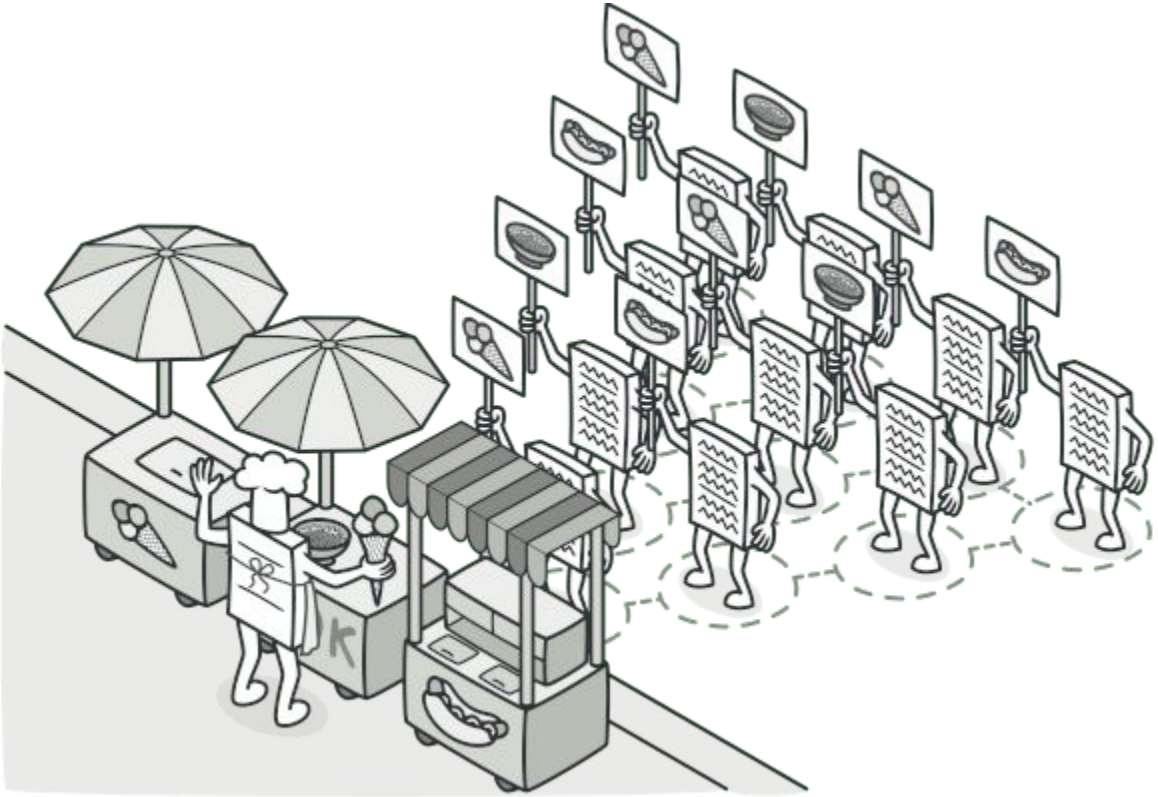
Benefits

- **Open/Closed Principle**. Introduce new observer classes without changing the observed class.
- Establish relations between objects at runtime.

Drawbacks

- Subscribers are notified in random order.
- Violates **Single Responsibility Principle** as the observed class needs to both maintain a list of observers and its original functionality.

Visitor



Visitor: Problem

```
1.  class AbstractNode{
2.    int _data;
3.  };
4.
5.  class RootNode {
6.    vector<AbstractNode*> _children;
7.  };
8.
9.  class Node : public AbstractNode {
10.   vector<AbstractNode*> _children;
11. };
12.
13. class FinalNode : public AbstractNode {
14. };
```

- We have a complex datastructure consisting of multiple interconnected classes.
- We want to add functionality that goes through the entire datastructure.

  Example: print function,
  export to XML function,
  …

Visitor: Solution

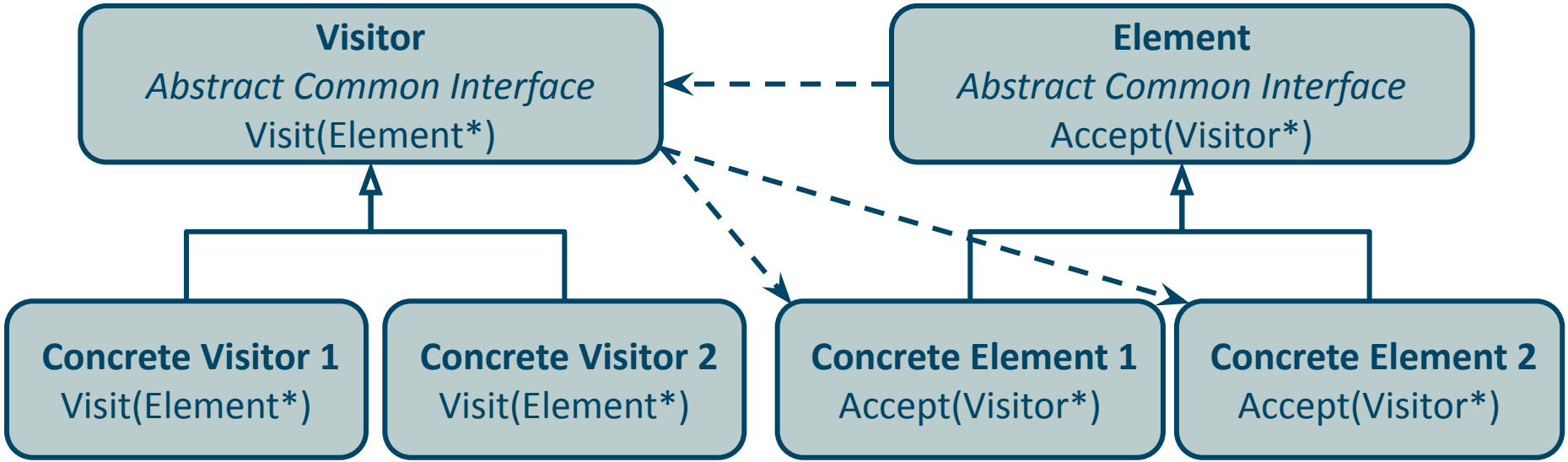
```
1.  class Visitor {
2.  public:
3.    virtual void visitRootNode(const RootNode * n) = 0;
4.    virtual void visitNode(const Node * n) = 0;
5.    virtual void visitFinalNode(const FinalNode * n) = 0;
6.  };
7.
8.  class TreePrintVisitor : public Visitor { ... };
9.  class TreeToXMLExportVisitor : public Visitor { ... };
```

Add an accept function to the datastructure that takes a visitor.

→ Implement a visitor for each functionality you need.

```
1.  class AbstractNode{
2.      int _data;
3.    public:
4.      virtual void accept(Visitor* v) const = 0;
5.  };
6.
7.  class RootNode {
8.      vector<AbstractNode*> _children;
9.    public:
10.     void accept(Visitor* v) const override {
11.       v->visitRootNode(this); };
12. };
13.
14. class Node : public AbstractNode {
15.     vector<AbstractNode*> _children;
16.   public:
17.     void accept(Visitor* v) const override {
18.       v->visitNode(this); };
19. };
```

Visitor: Structure



Visitor: Benefits & Drawbacks

Benefits

- **Open/Closed Principle.** Introduce new behavior that can work with objects of different classes without changing these classes.
- **Single Responsibility Principle.** Move multiple versions of the same behavior into the same class.
- A visitor object can accumulate useful information while working with various objects.

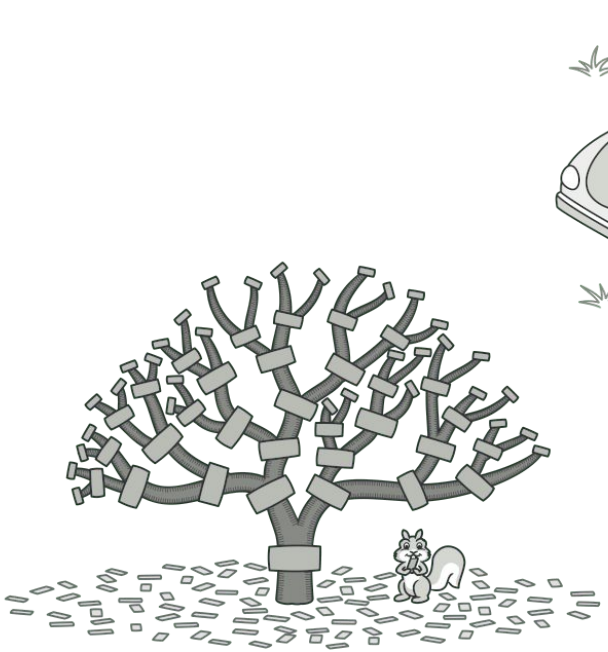
Drawbacks

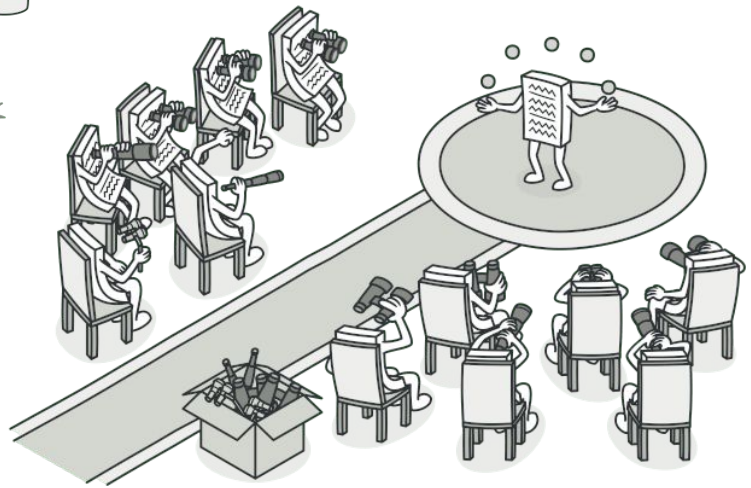
- Need to update all visitors each time a class gets added to the element hierarchy.
- Visitors might lack access to private fields.

Combining Patterns

Combine multiple patterns into more complex patterns.

Model-View-Controller





Combining the
Composite, Strategy, and Observer patterns

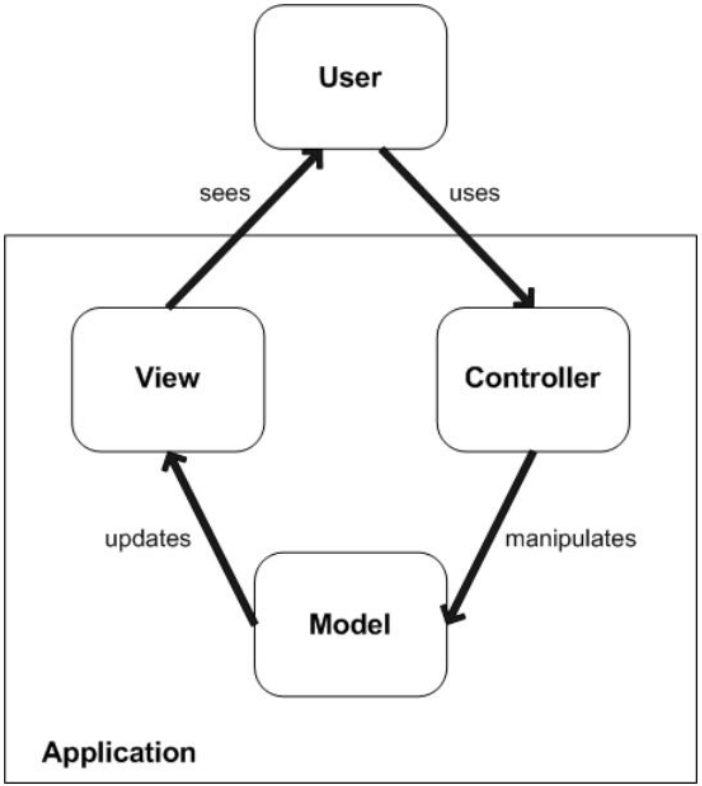
Model-View-Controller: Problem

- We want to develop a user interface for an application which allows the user to modify the data of the application.
- How can we implement this;
  - and keep the coupling between the graphical front-end and the database back-end low?
  - and satisfy the single responsibility principle?
    Example: graphical widgets should not be responsible for both visualizing data and manipulating data.
  - and satisfy the open/closed principle?
    Example: be able to adjust the UI without having to change the database.

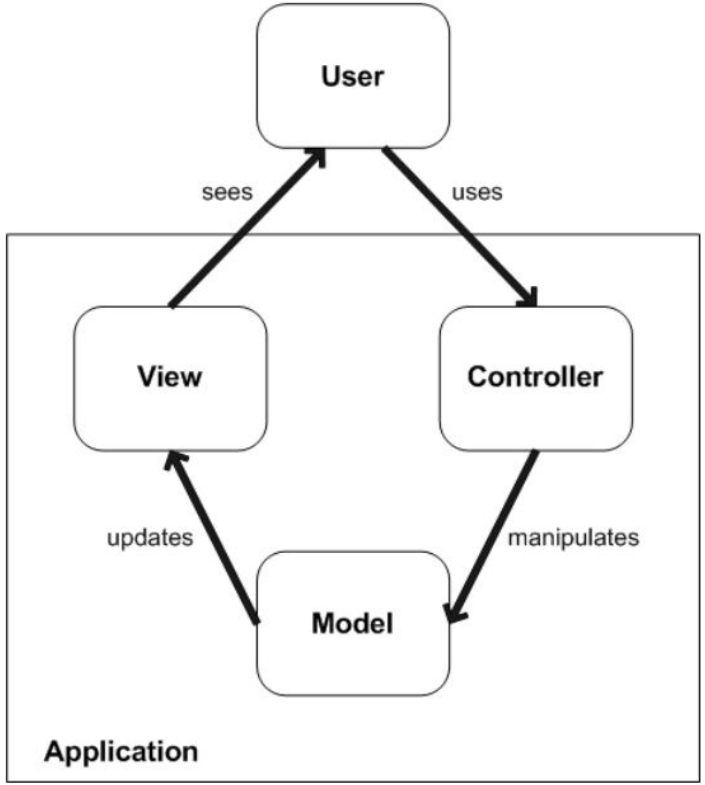
Model-View-Controller: Solution

Separate your application in three parts:

- The **Model** handles the back-end of your application. (database, simulation,...)
- The **View** handles the visual front-end of your application. (visualizing data, showing results,...)
- The **Controller** handles the user input of your application.



Model-View-Controller: Structure



MVC: Benefits & Drawbacks

Benefits

- **Open/Closed Principle.** Introduce new views or controllers without changing the model.
- **Single Responsibility Principle.** Separated the responsibility of showing data and handling user input.
- Low coupling between the front-end and the back-end.
- Can be implemented in a distributed system.

Drawbacks

- Increases the complexity and overhead in the application.

More patterns?

Free online resource: https://refactoring.guru/design-patterns

→ describes more patterns

→ detailed description of patterns, including examples